This problem will explore dark matter freeze-out in the early universe. We will derive approximate analytic expressions for the relic abundance of a heavy particle that was once in thermal equilibrium with the primordial plasma and froze out while non-relativistic. This is the classic "WIMP" (weakly interacting massive particle) scenario for dark matter production.

Consider a heavy dark matter particle χ (mass m_χ) and its antiparticle $\bar{\chi}$ in the early Universe. They can annihilate into a pair of light Standard Model (SM) particles $\psi_{\rm SM}, \bar{\psi}_{\rm SM}$,

$$\chi + \bar{\chi} \longleftrightarrow \psi_{\rm SM} + \bar{\psi}_{\rm SM}. \tag{0.1}$$

A few points to note:

- The light species $\psi_{\rm SM}, \bar{\psi}_{\rm SM}$ remain in full thermal and chemical equilibrium with the radiation bath at all times.
- There is no particle-antiparticle asymmetry in the χ sector: $n_\chi = n_{\bar{\chi}}$.
- The universe is radiation dominated during freeze-out.

Throughout, take the effective relativistic degrees of freedom in energy and entropy to be equal and approximately constant: $g_* \simeq g_{*S} \simeq 10$.

1 Equilibrium abundance of a non-relativistic species (2 points)

(a) Starting from the definition of the number density from the distribution function $f(p)$, show that the number density of a non-relativistic species χ in thermal and chemical equilibrium is approximately

$$n_\chi^{\rm eq}(T) \simeq g \left(\frac{m_\chi T}{2\pi}\right)^{3/2} e^{-m_\chi/T}, \tag{1.1}$$

where g is the number of internal degrees of freedom of χ (e.g. $g = 4$ for a Dirac fermion) and m_χ is its mass. Does this result depend on whether χ is a boson or a fermion?

We can start with the definition of number density:

$$n_\chi^{\rm eq} = g \int \frac{d^3p}{(2\pi)^3} f(p), \tag{1.2}$$

where $f(p)$ is the equilibrium distribution function. For a non-relativistic species, we can use the Maxwell-Boltzmann distribution:

$$f(p) = e^{-(E-\mu)/T} \simeq e^{-(m_\chi + p^2/(2m_\chi)-\mu)/T}, \tag{1.3}$$

where we have used $E \simeq m_\chi + p^2/(2m_\chi)$.

> Note that in the non-relativistic limit, the difference between bosons and fermions becomes negligible, so this result does not depend on the particle nature of χ.

Assuming chemical equilibrium with zero chemical potential ($\mu = 0$), we further simplify:

$$n_\chi^{\rm eq} = g \int \frac{d^3p}{(2\pi)^3} e^{-(m_\chi + p^2/(2m_\chi))/T} = g e^{-m_\chi/T} \int \frac{d^3p}{(2\pi)^3} e^{-p^2/(2m_\chi T)}. \tag{1.4}$$

The integral over momentum is a standard Gaussian integral:

$$\int d^3p\, e^{-p^2/(2m_\chi T)} = \int dp d\Omega\, p^2\, e^{-p^2/(2m_\chi T)} = (2\pi m_\chi T)^{3/2}. \tag{1.5}$$

The desired result is then obtained as

$$n_\chi^{\rm eq} = g e^{-m_\chi/T} \frac{(2\pi m_\chi T)^{3/2}}{(2\pi)^3} = g \left(\frac{m_\chi T}{2\pi}\right)^{3/2} e^{-m_\chi/T}. \tag{1.6}$$

(b) With the entropy density of the plasma,

$$s(T) = \frac{2\pi^2}{45}\, g_{*S}\, T^3, \tag{1.7}$$

we define the comoving equilibrium abundance

$$N_\chi^{\rm eq} \equiv \frac{n_\chi^{\rm eq}}{s}. \tag{1.8}$$

Derive an explicit expression for $N_\chi^{\rm eq}(T)$ in terms of $x \equiv m_\chi/T$, g, and g_{*S}, and show that for large x,

$$N_\chi^{\rm eq}(x) \propto x^{3/2} e^{-x}. \tag{1.9}$$

This is a trivial substitution:

$$N_\chi^{\rm eq}(T) = \frac{n_\chi^{\rm eq}}{s} = \frac{g\left(\frac{m_\chi T}{2\pi}\right)^{3/2} e^{-m_\chi/T}}{\frac{2\pi^2}{45} g_{*S} T^3}. \tag{1.10}$$

Simplifying, we get some rather unsettling fractions in our prefactor, but the general dependence on x is obtained

$$N_\chi^{\rm eq}(T) = \frac{45g}{2^{5/2}\pi^{7/2} g_{*S}} \left(\frac{m_\chi}{T}\right)^{3/2} e^{-m_\chi/T} = \frac{45g}{2^{5/2}\pi^{7/2} g_{*S}} x^{3/2} e^{-x}. \tag{1.11}$$

The point is that for large x (i.e., low temperatures), the entropy-weighted number of degrees of freedom g_{*S} is roughly constant, so we have recover

$$N_\chi^{\rm eq}(x) \propto x^{3/2} e^{-x}. \tag{1.12}$$

2 Simplifying the Boltzmann equation (4 points)

The Boltzmann equation for the number density of X in an expanding FRW Universe can be written as

$$\frac{dn_\chi}{dt} + 3Hn_\chi = -\langle \sigma v \rangle \left(n_\chi^2 - (n_\chi^{\rm eq})^2\right). \tag{2.1}$$

Recall, n_χ is the comoving number density of χ particles in the universe while $n_\chi^{\rm eq}$ is the comoving number density in thermal and chemical equilibrium. It is not true in general that $n_\chi = n_\chi^{\rm eq}$, but that is precisely what we want: n_χ should depart from $n_\chi^{\rm eq}$ as the universe expands and cools, such that eventually the annihilation rate becomes negligible and n_χ "freezes out" to a constant comoving abundance.

(a) Using the definition $N_\chi \equiv n_\chi/s$ and the fact that in adiabatic expansion $sa^3 = \text{const}$, show that the Boltzmann equation can be written as

$$\frac{dN_\chi}{dx} = -\frac{s}{Hx} \langle \sigma v \rangle \left(N_\chi^2 - (N_\chi^{\rm eq})^2\right). \tag{2.2}$$

where H is the Hubble parameter and $x \equiv m_\chi/T$. (Hint: $a \propto T^{-1}$ during radiation domination, so $\frac{dx}{dt} = Hx$.

Starting from the Boltzmann equation:

$$\frac{dn_\chi}{dt} + 3Hn_\chi = -\langle\sigma v\rangle \left(n_\chi^2 - (n_\chi^{\rm eq})^2\right). \tag{2.3}$$

We can express n_χ in terms of N_χ and s:

$$n_\chi = N_\chi s. \tag{2.4}$$

Taking the time derivative, we have:

$$\frac{dn_\chi}{dt} = s\frac{dN_\chi}{dt} + N_\chi\frac{ds}{dt}. \tag{2.5}$$

With adiabatic expansion, $sa^3 = \text{const}$, we find:

$$\frac{ds}{dt} = \frac{d(\text{const}\, a^{-3})}{dt} = -3\frac{\dot{a}}{a}a^3\text{const} = -3Hs. \tag{2.6}$$

Substituting this back into the expression for $\frac{dn_\chi}{dt}$ gives:

$$\frac{dn_\chi}{dt} = s\frac{dN_\chi}{dt} - 3HN_\chi s. \tag{2.7}$$

Plugging this into the Boltzmann equation, we get:

$$s\frac{dN_\chi}{dt} - 3HN_\chi s + 3HN_\chi s = -\langle\sigma v\rangle \left((N_\chi s)^2 - (N_\chi^{\rm eq} s)^2\right). \tag{2.8}$$

Simplifying, we have:

$$s\frac{dN_\chi}{dt} = -s^2\langle\sigma v\rangle \left(N_\chi^2 - (N_\chi^{\rm eq})^2\right) \implies \frac{dN_\chi}{dt} = -s\,\langle\sigma v\rangle \left(N_\chi^2 - (N_\chi^{\rm eq})^2\right). \tag{2.9}$$

To change variables from t to x, we use the relation $\frac{dx}{dt} = Hx$, which gives:

$$\frac{dN_\chi}{dx} = -\frac{s}{Hx}\langle\sigma v\rangle \left(N_\chi^2 - (N_\chi^{\rm eq})^2\right). \tag{2.10}$$

This is a rather welcome simplification. The equation is still quadratic in N_χ, but now the expansion of the universe is encapsulated in the prefactor $s/(Hx)$, and not in a separate linear term. The form of this ODE is called a *Riccati equation* and is not generally solvable in closed form. It can be transformed into a second-order linear ODE, but that is not particularly helpful here, but we will motivate what a solution to this equation looks like in the next sections.

(c) For a radiation-dominated Universe, the Hubble parameter is

$$H(T) = \sqrt{\frac{4\pi^3}{45}}\frac{\sqrt{g_*}\,T^2}{M_{\rm Pl}}, \tag{2.11}$$

with reduced Planck mass $M_{\rm Pl} \simeq 2.4 \times 10^{18}\,\text{GeV}$. Evaluate $s/(Hx)$ at general temperature and then at $T = m_\chi$ (i.e. $x = 1$) to define a constant

$$\lambda \equiv \frac{s(m_\chi)\,\langle\sigma v\rangle}{H(m_\chi)}. \tag{2.12}$$

Show that Eq. (2.2) can be written in the "Riccati" form

$$\frac{dN_\chi}{dx} = -\frac{\lambda}{x^2}\left(N_\chi^2 - (N_\chi^{\rm eq})^2\right). \tag{2.13}$$

Give λ explicitly in terms of g_*, g_{*S}, m_χ, $M_{\rm Pl}$, and $\langle\sigma v\rangle$.

First, let me say what the point of this is: we want a ODE that tracks the evolution of N_χ in a relative way. So, by evaluating $s/(Hx)$ at $T = m_\chi$, we can factor out all the dimensional quantities into a single parameter λ that sets the scale of the interaction rate relative to the expansion rate at the time when $T = m_\chi$. The point is that we will argue the λ is roughly constant during freeze-out ($g_* \simeq g_{*S} \simeq 10$ is roughly constant), so the only time dependence in the ODE will be in the $1/x^2$ prefactor and in $N_\chi^{\rm eq}(x)$. To that end, we write

$$\frac{s}{Hx} = \frac{\frac{2\pi^2}{45} g_{*S} T^3}{\sqrt{\frac{4\pi^3}{45}}\frac{\sqrt{g_*}T^2}{M_{\rm Pl}}x} = \frac{2\pi^2}{45} g_{*S}\frac{T^3 M_{\rm Pl}}{\sqrt{\frac{4\pi^3}{45}}\sqrt{g_*}T^2 x}. \tag{2.14}$$

Simplifying, we get:

$$\frac{s}{Hx} = \sqrt{\frac{\pi}{45}}\frac{g_{*S}}{\sqrt{g_*}}\frac{M_{\rm Pl}T}{x}. \tag{2.15}$$

Evaluating this at $T = m_\chi$ (i.e., $x = 1$), we find:

$$\lambda = \frac{s(m_\chi)\langle\sigma v\rangle}{H(m_\chi)} = \sqrt{\frac{\pi}{45}}\frac{g_{*S}}{\sqrt{g_*}}M_{\rm Pl}m_\chi\langle\sigma v\rangle. \tag{2.16}$$

Substituting this back into Eq. (2.2), we obtain the Riccati form:

$$\frac{dN_\chi}{dx} = -\frac{\lambda}{x^2}\left(N_\chi^2 - (N_\chi^{\rm eq})^2\right). \tag{2.17}$$

3 Approximate analytic solution (2 points)

The Riccati equation (2.13) cannot be solved in closed form with the exact equilibrium abundance, but it admits a standard approximate solution using the *freeze-out approximation.*

At early times (small x), interactions are rapid and $N_\chi \approx N_\chi^{\rm eq}$. In this regime the interaction rate per particle $\Gamma \equiv n_\chi^{\rm eq}\langle\sigma v\rangle$ satisfies $\Gamma \gg H$.

(a) Define the freeze-out point x_f by the condition

$$\Gamma(x_f) \approx H(x_f) \quad \Rightarrow \quad n_\chi^{\rm eq}(x_f)\langle\sigma v\rangle \approx H(x_f). \tag{3.1}$$

Using your expressions from Problems 1 and 2, derive an implicit equation determining x_f:

$$x_f \simeq \ln\left[c(c+2)\,\frac{g}{\sqrt{g_*}}\,\frac{m_\chi M_{\rm Pl}\langle\sigma v\rangle}{x_f^{1/2}}\right], \tag{3.2}$$

where c is an $\mathcal{O}(1)$ numerical constant introduced to refine the criterion (you may take $c \approx 1 - 1.5$).
(*Hint:* Start from Eq. (3.1), plug in the explicit forms of $n_\chi^{\rm eq}$ and H, and isolate logarithms of x_f.)

> Starting from the freeze-out condition:
>
> $$n_\chi^{\rm eq}(x_f)\langle\sigma v\rangle \approx H(x_f). \tag{3.3}$$
>
> Substituting the expressions for $n_\chi^{\rm eq}$ and H, we have:
>
> $$g\left(\frac{m_\chi T_f}{2\pi}\right)^{3/2} e^{-m_\chi/T_f}\langle\sigma v\rangle \approx \sqrt{\frac{4\pi^3}{45}}\,\frac{\sqrt{g_*}T_f^2}{M_{\rm Pl}}, \tag{3.4}$$
>
> where $T_f = m_\chi/x_f$. Rearranging, we get:
>
> $$e^{-x_f} \approx \sqrt{\frac{4\pi^3}{45}}\,\frac{\sqrt{g_*}}{g}\left(\frac{2\pi}{m_\chi T_f}\right)^{3/2}\frac{T_f^2}{M_{\rm Pl}\langle\sigma v\rangle}. \tag{3.5}$$
>
> Taking the natural logarithm of both sides, we find:
>
> $$-x_f \approx \ln\left[\sqrt{\frac{4\pi^3}{45}}\,\frac{\sqrt{g_*}}{g}(2\pi)^{3/2}m_\chi^{-3/2}T_f^{-1/2}M_{\rm Pl}^{-1}\langle\sigma v\rangle^{-1}\right]. \tag{3.6}$$
>
> Rearranging gives:
>
> $$x_f \simeq \ln\left[c(c+2)\,\frac{g}{\sqrt{g_*}}\,\frac{m_\chi M_{\rm Pl}\langle\sigma v\rangle}{x_f^{1/2}}\right], \tag{3.7}$$
>
> where c is an $\mathcal{O}(1)$ numerical constant introduced to refine the criterion of freeze-out (the decoupling point is not instantaneous at $\Gamma = H$).
>
> > Note that this is an implicit equation for x_f since it appears on both sides of the equation. At this point, you can plug in some Weak-interaction cross sections values for $\langle\sigma\rangle \simeq G_F^2 m_\chi^2 v \sim 10^{-9}\,\mathrm{GeV}^{-2}$ where we took for example $m_\chi \simeq 10$ GeV and $v \sim 0.1$. In this case, you will find $x_f \approx 20$, which is consistent with typical freeze-out temperatures for WIMPs: they are non-relativistic!

(c) After freeze-out, $N_\chi^{\rm eq}$ becomes exponentially small, and the Boltzmann equation (2.13) simplifies to

$$\frac{dN_\chi}{dx} \approx -\frac{\lambda}{x^2} N_\chi^2, \qquad (x \gtrsim x_f). \tag{3.8}$$

Solve this differential equation analytically for $N_\chi(x)$ for $x \geq x_f$, imposing the matching condition $N_\chi(x_f) \approx N_\chi^{\rm eq}(x_f)$, and obtain the asymptotic abundance

$$N_\chi(\infty) \equiv N_\infty \approx \frac{x_f}{\lambda}. \tag{3.9}$$

Why is N_∞ inversely proportional to the annihilation cross section $\langle \sigma v \rangle$?

> Starting from the simplified Boltzmann equation after freeze-out:
>
> $$\frac{dN_\chi}{dx} \approx -\frac{\lambda}{x^2} N_\chi^2. \tag{3.10}$$
>
> We can separate variables, assume λ is roughly constant, and integrate
>
> $$\int_{N_\chi(x_f)}^{N_\chi(x)} \frac{dN_\chi}{N_\chi^2} = -\lambda \int_{x_f}^{x} \frac{dx'}{x'^2}. \tag{3.11}$$
>
> Evaluating the integrals, we have:
>
> $$-\frac{1}{N_\chi(x)} + \frac{1}{N_\chi(x_f)} = \lambda \left(\frac{1}{x_f} - \frac{1}{x} \right). \tag{3.12}$$
>
> Taking the limit as $x \to \infty$, we find:
>
> $$-\frac{1}{N_\infty} + \frac{1}{N_\chi(x_f)} = \frac{\lambda}{x_f}. \tag{3.13}$$
>
> Rearranging gives:
>
> $$N_\infty = \frac{x_f}{\lambda + x_f / N_\chi(x_f)}. \tag{3.14}$$
>
> Since $N_\chi(x_f) \approx N_\chi^{\rm eq}(x_f)$ is typically much larger than x_f/λ, we can approximate:
>
> $$N_\infty \approx \frac{x_f}{\lambda}. \tag{3.15}$$
>
> > The asymptotic abundance N_∞ is inversely proportional to the annihilation cross section $\langle \sigma v \rangle$ because a larger cross section leads to more efficient annihilation of χ particles before freeze-out, resulting in a lower final abundance.

4 Relic density and the "WIMP miracle" (2 points)

(a) The present-day dark matter density in units of the critical density is

$$\Omega_\chi h^2 = \frac{\rho_\chi}{\rho_c} h^2 \simeq \frac{m_\chi n_\chi^0}{\rho_c / h^2} = \frac{m_\chi s_0 N_\infty}{\rho_c / h^2}, \tag{4.1}$$

where $s_0 \simeq 2890\,\mathrm{cm}^{-3}$ is today's entropy density and $\rho_c/h^2 \simeq 1.05\times 10^{-5}\,\mathrm{GeV\,cm}^{-3}$ (h is the dimensionless Hubble parameter). Note we used $n_\chi^0 = s_0 N_\infty$ since N_χ is comoving and thus constant after freeze-out. Using your result for N_∞, show that

$$\Omega_\chi h^2 \approx \frac{0.1\,\mathrm{pb}}{\langle\sigma v\rangle} \tag{4.2}$$

up to $\mathcal{O}(1)$ factors and logarithms, where $1\,\mathrm{pb} = 10^{-36}\,\mathrm{cm}^2$.
(Hint: Convert $\langle\sigma v\rangle$ from GeV^{-2} to cm^3/s as needed.)

(b) Take the observed dark matter abundance to be $\Omega_{\rm DM}h^2 \simeq 0.12$. For a WIMP with mass $m_\chi = 100\,\mathrm{GeV}$, estimate the required annihilation cross section $\langle\sigma v\rangle$. Compare your answer to a "typical" weak-interaction cross section. Explain why this is often referred to as the *WIMP miracle*.

> Let tackle both items (a) and (b). Through a series of substitutions, we have:
>
> $$\Omega_\chi h^2 = \frac{m_\chi s_0 N_\infty}{\rho_c/h^2} = \frac{m_\chi s_0 x_f}{\lambda \rho_c/h^2}. \tag{4.3}$$
>
> Substituting the expression for λ, we get:
>
> $$\Omega_\chi h^2 = \frac{m_\chi s_0 x_f}{\sqrt{\frac{\pi}{45}}\frac{g_{*S}}{\sqrt{g_*}} M_{\rm Pl} m_\chi \langle\sigma v\rangle \rho_c/h^2} = \frac{s_0 x_f \sqrt{g_*}}{\sqrt{\frac{\pi}{45}} g_{*S} M_{\rm Pl}\langle\sigma v\rangle \rho_c/h^2}. \tag{4.4}$$
>
> Plugging in the numerical values, we find:
>
> $$\Omega_\chi h^2 \approx \frac{0.1\,\mathrm{pb}}{\langle\sigma v\rangle}. \tag{4.5}$$
>
> For a WIMP with mass $m_\chi = 100\,\mathrm{GeV}$ and the observed dark matter abundance $\Omega_{\rm DM}h^2 \simeq 0.12$, we find:
>
> $$\langle\sigma v\rangle \approx \frac{0.1\,\mathrm{pb}}{0.12} \approx 0.83\,\mathrm{pb}. \tag{4.6}$$
>
> This value is remarkably close to the typical weak-interaction cross section, which is on the order of 1 pb. This coincidence is often referred to as the *WIMP miracle*, as it suggests that particles with weak-scale interactions naturally yield the correct relic abundance of dark matter observed in the universe.

5 Numerical solution (2 bonus points)

This problem is optional and intended as a programming exercise (e.g. in Python, Mathematica, or C++).

(a) Starting from the Riccati equation

$$\frac{dN_\chi}{dx} = -\frac{\lambda}{x^2}\left(N_\chi^2 - (N_\chi^{\rm eq})^2\right), \tag{5.1}$$

implement a numerical ODE solver to evolve $N_\chi(x)$ from an initial point $x_{\text{init}} \ll 1$ (e.g. $x_{\text{init}} = 1$) to a large final value $x_{\text{final}} \gg x_f$ (e.g. $x_{\text{final}} = 1000$).

(b) Compare your numerical solution for $N_\chi(x)$ with:

1) the equilibrium abundance $N_\chi^{\text{eq}}(x)$, and
2) the analytic freeze-out approximation from Problems 3–4.

Plot all three curves on the same graph and identify the freeze-out epoch.

You can also vary $\langle \sigma v \rangle$ over at least two orders of magnitude and study how the final relic abundance changes. Confirm the inverse proportionality $\Omega_\chi \propto 1/\langle \sigma v \rangle$ to good accuracy.